NIWE ERP

Administrator Manual

Install, set up and look after NIWE ERP

Version 1.27.0

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1 About this manual

1.1 Who it is for

This is the **Administrator Manual** for NIWE ERP (the NIWE POS/ERP system). It is written for the person who installs, configures and looks after the system: the business owner, the IT person, or the office manager who holds the **Admin** account. Day-to-day instructions for cashiers, accountants, store keepers and other staff are in the separate **User Manual** and in the **Role quick-start cards**.

You do not need to be a programmer to follow most of this manual. Chapters 2 and 12 (installation and upgrades) are technical and are meant for whoever manages the server.

1.2 How the manual is organised

Chapter	What it covers
2. Installing and first start	Requirements, database, starting the system, first sign-in
3. Licence and modules	Installation id, licence file, activating and requesting modules
4. Users, roles and permissions	Staff accounts, custom roles, per-person approval rights, branch access
5. Business set-up	Business details, currency, taxes, payment methods, branches, warehouses
6. Accounting set-up	Chart of accounts, cost centers, controls, opening balances
7. Module settings	Where each module keeps its own settings
8. Tills, printers and devices	Thermal printing, cash drawer, customer display, binding a register to a PC
9. E-mail, alerts and scheduled reports	SMTP, business alerts, scheduled reports, daily stock jobs
10. Backup, restore, import and export	Nightly backup, restore, bulk import with check-first, year-end archive
11. Logs and monitoring	Audit log, sign-in log, server logs
12. Upgrades and migrations	Installing a new version, database migrations
13. Security checklist	What to change before going live
14. Troubleshooting	Common problems and what to do

1.3 Conventions

- Menu paths are written like **Settings > Modules**. The words are exactly as they appear on screen.
- Buttons and fields are written in **bold**.
- Boxes marked *Note*, *Tip* and *Warning* call attention to things that matter. A warning describes something that cannot easily be undone.

1.4 What is new in this edition

This edition covers everything up to **version 1.27.0**. The main administrator-facing additions since version 1.12 are:

- **1.12** Thermal (Epson ESC/POS) printing, cash drawer and COM customer display.
- **1.13** A register can be tied to one known device.
- **1.14** Server logs and the sign-in log; more actions in the audit log; currency follows the chart of accounts (RWF for Rwanda, XAF for Chad).
- **1.15 to 1.22** Improvements to Work Orders, Time Tracking, Booking, Installments, Subscriptions, Help Desk, Expenses and Assets, including new approval permissions (*Approve large expenses*, *Approve large asset disposals*) and new reports available to scheduled reports.
- **1.22.1 to 1.22.5** Fixes and hardening: subscription month-end billing, first-start administrator on a real installation, no password or secret kept in the shipped files, the server logs written next to the jar, and a printable PDF of the modules list.
- **1.23** French. Every user can switch between English and Français from the top bar or the sign-in page, and the business has a default language (Settings > System settings) that follows the chart of accounts: Chad starts in French, Rwanda in English. The core screens are translated: sign-in, menus, dashboard, POS, Sales, Purchases, Inventory, Customers, Accounting and Settings.
- **1.24** French for the other modules: Restaurant, Hotel, Manufacturing, Subscriptions, Installments, Help Desk, HR and Payroll, Time tracking, Work orders, Booking, Law firm, Real estate, Property management, Assets, Gift cards, Loyalty, Promotions, the client portal, My profile and the Invoice list. A phrase without a French version shows in English. Ask a French speaker to read the wording before customers use it.
- **1.25** French for the messages written in the program code (error messages, "Are you sure?" questions, notices), for lists such as status filters, weekdays and permission names, and for the titles and column headings of every report. A fix makes texts that carry a value (for example "3 days left") show correctly in English. Server messages, e-mails and printed PDF documents were still English in this version.
- **1.26** French for the texts the server writes: printed documents (invoices, quotations, receipts, statements, delivery and goods-received notes, payslips, hotel bills, law-firm letters, instalment agreements, the modules list and more), the e-mails sent to customers, tenants, owners, guests, clients and staff, and the messages the server returns (for example "Customer not found"). They follow the **business language** in Settings > System settings, not the language a user picked on their own screen. Dates and amounts are written the French way (5 mars 2026, 1 234,50). Audit trail entries and names already stored in the database stay as they were written. Ask a French speaker to read the wording before customers receive it.
- **1.26.1** Fix: "Failed to create backup directory ./backups" when the server runs as a service. Backups now go to a **backups** folder next to the jar by default.
- **1.26.2** The log files go into a folder named **logs** next to the jar (instead of the jar's folder itself).
- **1.27.0** New optional module **Online Store**: a public shop at `/shop` where customers order for delivery or pick-up and pay in cash. See *Online Store settings* in chapter 7 and the public-side notes in chapter 13.

2 Installing and first start

2.1 What the system is made of

NIWE ERP has two parts that run together:

- The **backend**, a Java 21 / Spring Boot application, which holds all the business rules and talks to the database. It listens on port **8080**.
- The **frontend**, an Angular 18 web application that staff open in a browser. In development it is served on port **4200**; in production it is normally built into static files and served by a web server.
- A **PostgreSQL** database that stores everything. The database structure is created and upgraded automatically by **Flyway**.

Staff only need a modern browser (Chrome or Edge is recommended; thermal printing needs one of these).

2.2 Requirements

- Java 21 and Maven (to build or run the backend) or the packaged backend you were given.
- Node.js and npm (to build the frontend).
- PostgreSQL, with an empty database.
- For thermal printing from a till PC: Chrome or Edge, opened on **https** or on **localhost** (see chapter 8).

2.3 First installation

1. **Create the database.** For example: `createdb pos_app`.
2. **Set the database connection.** Give the backend the address, user and password of your database through environment variables: `DB_URL` (for example `jdbc:postgresql://localhost:5432/pos_app`), `DB_USER` and `DB_PASSWORD`, and set `JWT_SECRET` to a long random value. Alternatively put the same settings (`spring.datasource.url`, `.username`, `.password`, `app.jwt.secret`) in a private file `config/application.properties` next to where the backend is started. No password is kept in the shipped files; from version 1.22.3 the shipped files hold only harmless defaults.
3. **Start the backend.** On the very first start Flyway creates all tables (migration V1, the baseline) and the licence tables (V2) and then applies every later migration. This can take a minute. Wait for the message *Started PosApplication*.
4. **Start the frontend.** In development: `npm install` then `npm start`, and open `http://localhost:4200`. If the backend is somewhere other than `localhost:8080`, change `API_BASE` in `frontend/src/app/core/constants.ts` before building, and add the frontend's address to `app.cors.allowed-origins` in the backend properties.
5. **Sign in** as `admin` with the password `admin123` (when demo data is on, the default). On a real installation without demo data, see *A real installation without demo data* below.

> **Warning:** The demo accounts (`admin` / `admin123`, `cashier` / `cashier123`, with terminal PINs 1234 and 5678) are only meant for the first start. **Change the admin password immediately**, and set

> `JWT_SECRET` (the `app.jwt.secret` setting) to a long random value of your own. Chapter 13 has the full checklist.

2.4 Demo data

On the first start the system adds demo data: an admin and a cashier, four categories, eight barcoded products, a walk-in customer, the payment methods, a standard tax rate and one register. This is helpful for learning or for a first look. It is added only once.

2.4.1 A real installation without demo data

For a real business, start the very first time with `app.seed-demo-data=false`. With no demo data and an empty database, the system creates one administrator, `admin`, so that you can sign in:

- If you set `app.initial-admin-password=` to a password of your own, that is the password.
- If you leave it empty, the system makes a random password and writes it **once** to the log, in a line that starts with *FIRST START* (chapter 11). Copy it, sign in, and change it at once under your profile.

This happens only when there are no users at all, so a restart never changes your password or creates another account. Without the demo data there are also no payment methods, taxes or register yet: create them in **Settings** (steps 5 and 11 below). Do not use the demo accounts on a server that real people can reach.

2.5 Order of set-up for a new business

A sensible order that avoids rework:

1. Change the admin password (**Staff**, then your own account) and the JWT secret.
2. Fill in **Settings > Business details**.
3. Install the licence and activate modules (**Settings > Modules**, chapter 3).
4. Set **System settings** (currency, time zone, date format, discount mode).
5. Create **Taxes** and **Payment methods**.
6. Create **Branches** and **Warehouses** (if you have more than the head office).
7. Choose the **chart of accounts** and set the accounting controls (chapter 6).
8. Create **Roles** and **Staff** accounts (chapter 4).
9. Set up **E-mail settings** (chapter 9), then the settings of each module you use (chapter 7).
10. Import products, customers, suppliers and opening balances (chapter 10), checking each file first.
11. Set up registers, printers and devices (chapter 8).
12. Make a first manual backup (**Settings > Backup & Data**).

2.6 Where the system keeps its files

Besides the database, the backend writes its log files (chapter 11) and keeps uploaded documents and photos in its storage folder. When the backend runs as a jar, for example as a service, the log files are written to a folder named **logs** next to the jar (`niwe-erp.log`, `niwe-erp-error.log`, `niwe-access.log`) and the database backups to a folder named **backups** next to the jar; both are created when missing, so the account that

runs the service must be allowed to write in the jar's folder. Include both the database backup and the storage folder in your own server backup routine.

3 Licence and modules

3.1 How licensing works

Which modules a business may use is decided by a **licence file** that NIWE signs for that one installation. A module can be used only when it is **in the licence** *and* an administrator has **activated** it.

- **Core modules** are always active: **Sales, Purchases and Accounting**.
- Products and stock, customers, suppliers, staff and settings are shared by every module and are always available.
- All other modules (for example Hotel, Restaurant, Online Store, Payroll, Assets) need to be in the licence and then activated.

The licence file has the extension `.niwelic`. It is signed with a digital signature (Ed25519) and checked inside the application, so it cannot be edited by hand. It is checked offline: no internet connection is needed.

3.2 The installation id

Every installation gets a random **installation id** the first time it starts. It is stored in the database, so it travels with backups. A licence is issued for exactly this id and will not work on another installation.

You find it on **Settings > Modules**, in the *License* section, next to a **Copy** button.

3.3 Getting and installing a licence

1. Open **Settings > Modules** and press **Copy** next to the installation id.
2. Send the id to NIWE together with the list of modules the business bought.
3. NIWE sends back a `.niwelic` file.
4. On **Settings > Modules**, upload the file. The message *License installed* appears and the licence is listed under *Installed licenses* (number, issued date, expiry date, modules, who installed it).
5. Activate the modules the business needs (next section).

The screen shows who the licence is issued to, its number, issue date and expiry date (or *no expiry date*).

3.4 Activating modules

Each module card on **Settings > Modules** shows a state:

State on screen	Meaning
Core / Always on	Sales, Purchases, Accounting. Nothing to do.
In your license	Licensed but not yet switched on. Press **Activate**.
Active	In use. It cannot be turned off again.
Locked (Request)	Not in your licence. You may press **Request**.
Requested	A request was sent; it is waiting for NIWE. You may **Withdraw request**.

State on screen	Meaning
License expired	The module was in a licence that has expired.

> **Warning:** Activation is permanent. Once a module is activated it cannot be deactivated. Activate only what the business really needs; a module that is not activated is hidden from every menu.

Some modules work together. When you activate one, the system may activate another that it depends on and tells you so ("Also activated, because it works together with..."). The module cards show *Works together with* and *Needed by* lines.

After you activate **Hotel**, the screen reminds you to set it up in this order: first room types (name and nightly rate), then rooms, then the hotel settings (accounting mode, tax, times). The *Charge to room* payment method then becomes available in the POS and the Restaurant.

3.5 Printing the list of modules

Press **Print PDF** at the top of **Settings > Modules** to get an A4 PDF of every module with what it does, the modules it works together with, and its state in this installation. The top of the page shows who the licence is issued to, its number, expiry date and the installation id, followed by a count of core, active, available and locked modules. This is handy to send to NIWE with a request, or to keep with the business records.

3.6 Requesting a module you do not have

On a locked module press **Request**, optionally add a note (why you need it, how many users, when) and press **Send request**.

- If e-mail is set up (**Settings > Email settings**) and the setting `app.license.vendor-email` has an address, the request is e-mailed to NIWE with the installation id and the business name.
- Otherwise the screen shows the text so you can send it by hand.

> **Note:** A request unlocks nothing. The module becomes available only when you install a new licence file that contains it.

3.7 Expiry

A licence may have an expiry date. During the **last 30 days** an administrator sees a warning on **Settings > Modules** ("The license expires on ... (n days). Ask NIWE for a renewal in good time."). After the date:

- Non-core modules **lock**; Sales, Purchases and Accounting keep working.
- **Nothing is deleted.** Your data is untouched.
- Installing a renewed licence restores access immediately.

3.8 What a locked module does

It is greyed out on **Settings > Modules** and hidden from menus. The server refuses calls to its own screens' data (HTTP 403, "X is not part of your license"), its public pages answer "not available", and its morning jobs are skipped.

Reports, the dashboard, products, customers, invoices and stock are shared and stay open. Scheduled e-mail reports are not stopped when a module is locked.

3.9 For developers only

The setting `app.license.dev-all-modules=true`, together with the `dev` profile, makes every module count as licensed. It is ignored unless the `dev` profile is active.

> **Warning:** Never start a customer installation with the `dev` profile. If the screen shows "The developer switch is on: every module counts as licensed", the installation is running in development mode and must be restarted normally.

4 Users, roles and permissions

4.1 The three layers of access

NIWE ERP controls what each person can do in three layers. Understanding them makes user set-up easy.

1. **The base role** of the account: **Admin** or **Cashier**. An Admin can do everything. A Cashier does day-to-day work (the till, and anything the next two layers add).
2. **A custom role** (optional, one per person), built under **Staff > Roles**. It adds *access areas* to a Cashier account, each set to *No access*, *View only* or *Manage*. Examples: Manager, Accountant, Store keeper.
3. **Extra permissions** (optional, per person): individual approval rights, such as *Approve expenses* or *Override the store-wide discount cap*, that you grant one by one without making the person an Admin.

On top of this, a Cashier can be limited to specific **branches**.

> **Note:** The server enforces all of this. Hiding a menu item is only a convenience; a person without access is also refused by the server if they try to reach the data another way.

4.2 Creating a staff account

Go to **Staff** and press the button to add a new staff member. The form has these fields:

Field	What to enter
Username	The sign-in name.
Password	The initial password. Ask the person to change it after the first sign-in.
Full name	As it should appear in reports and on approvals.
Email	Optional. Needed only so the person can use **Forgot password?** on the sign-in screen.
Role	**Cashier** or **Admin**.
Custom role	*None*, *plain cashier*, or a role you built under Staff > Roles. Gives the access of that role on top of normal cashier work. Admins always have full access.
Active	Untick to stop the person signing in without deleting the account.
Extra permissions	Individual approval rights (below).
Branch access	Visible when you have more than one branch and the role is Cashier.

Press **Save**. To stop someone from working, untick **Active** rather than deleting the account: their name stays on past sales, approvals and audit entries.

> **Tip:** Give every person their own account. Shared accounts make the audit log and the cashier reports useless.

4.2.1 Terminal PINs

Cashiers open the sales terminal and switch cashier by typing a **PIN**. Each person's PIN identifies them, so two cashiers can ring sales on one shift without logging out. On the **Staff** list, each row shows whether a PIN is **Set** or **Not set**, with a **Set PIN** / **Change PIN** button. A PIN must be 4 to 6 digits and unique among active staff. The demo accounts have PINs 1234 (admin) and 5678 (cashier); change them.

4.2.2 Branch access

If the business has several branches, tick the branches a Cashier may work in. The person will then see and work only those branches' tills, sales and approvals. **Leave all unticked for access to every branch.** Admins are never limited.

4.3 Custom roles

Open **Staff > Roles** and press **New role**.

- **Role name**: for example *Manager*, *Accountant*, *Store keeper*.
- **Description**: optional.
- **Active**: an inactive role gives its staff no extra access until it is switched back on.
- **What this role can do**: one line per access area, each set to **No access**, **View only** or **Manage**. *View* opens the screens and lists read-only. *Manage* also lets the person create and change things.

Two ready-made roles, **Manager** and **Accountant**, exist for you to use or copy.

4.3.1 The access areas

Area	Typical screens it opens
Sales and customers	Customers, quotations, sale orders, price lists, promotions, gift cards, loyalty, sale-order returns, shifts, cash movements, POS voids and returns, registers; customer payments and collections
Billing and payments	Invoices, recurring invoices, credit notes, refunds, bills, supplier payments, installments; customer payments and collections; turning approved time into an invoice; recurring expenses
Inventory and purchasing	Products, categories, brands, warehouses, suppliers, purchase requisitions and orders, purchase returns, stock adjustments, counts, transfers, reorder, packaging
Operations	Manufacturing, Restaurant, Subscriptions, Help Desk, Time tracking, Work orders, Booking, Law firm, Real estate, Property management; devices
Accounting	Chart of accounts, cost centers, journal entries, controls, incomes, expenses, **Assets**
Hotel	The whole Hotel module
Human resources	Employees, documents, onboarding, attendance, leave, org chart
Payroll and contracts	Contracts and payroll (these carry salary data, so they have

Area	Typical screens it opens
	an area of their own)
Reports and dashboard	The dashboard and the reports of each module (view only). Some salary reports stay locked even here.

A report screen opens to a person who has *View* on **Reports and dashboard** *or* on the module's own area, so you can let a store keeper see inventory reports without opening the whole reports area.

Only one custom role can be given to a person. If someone needs the powers of two roles, make one combined role.

4.3.2 Screens that cannot be delegated

The following are always **Admin only**, whatever roles you build: **Staff and Roles, Settings** (Modules, Payment methods, Taxes, Business details, System settings, Email settings, Scheduled reports, Apps, Audit Log, Sign-in log, Server logs), **Backup & Data** including import and export, and every module's own **Settings** page. Roles never open them.

4.4 Per-person approval permissions

Some actions need a second pair of eyes: a manager approving a write-off, for example. Instead of making that manager an Admin, tick the specific rights under **Extra permissions** on their staff record. An Admin already has all of them.

Permission	What it allows
Confirm/reject Restaurant wastage	Review a logged wastage entry: apply its stock and accounting effect, or discard it
Approve general Inventory stock adjustments	Review a logged stock adjustment
Approve Purchase requisitions	Approve or reject a requisition awaiting approval
Override the store-wide discount cap	Apply a discount at checkout and exceed the configured maximum discount
Approve/reject Expenses	Approve or reject an expense awaiting approval, and void an approved one
Approve large Expenses	Approve an expense above the approval limit set in Expenses > Settings
Approve large asset disposals	Dispose of an asset whose book value is above the limit set in Assets > Settings, and undo a disposal
Approve/reject Stock transfers	Review a logged transfer between warehouses
Approve voids and returns at the till	Approve a void or a large return by typing a PIN, without any other manager right
Approve price changes at the till	Approve a hand-changed price by typing a PIN
Approve big paid-outs and cash corrections	Approve a large paid-out, and cancel a mistaken cash movement

> **Tip:** The three "at the till" permissions let you give a shift supervisor just enough power to approve exceptions by typing their PIN, without opening any back-office screen to them.

4.5 Good practice

- Keep the number of Admins small (two is typical: the owner and one deputy).
- Review the staff list every quarter; untick **Active** for leavers on the day they go.
- Use the **Audit Log** (chapter 11) to see who approved what.
- Prefer a role plus a few extra permissions to making people Admin.

5 Business set-up

This chapter covers the shared settings that every module uses. All are under **Settings** unless stated.

5.1 Business details

Settings > Business details holds the name, address, phone, e-mail and **Tax ID** of the business, and its **logo**. These print on receipts, invoices, quotations, statements and e-mails, and the business name is included in module requests to NIWE. Fill it in before you issue the first document.

5.2 System settings

Settings > System settings (press **Save** at the bottom; *System settings saved* confirms):

Setting	Notes
Currency	Pick the currency. The **Currency symbol** is filled in from it; override it if you would rather show a code (for example "UGX").
Business language	**English** or **Français**. This is the language everyone sees until they pick their own with the language selector on the top bar or the sign-in page. It is set for you when you choose the Chad chart of accounts (Français) and stays English for Rwanda, but only while it has not been changed. From version 1.26 it also decides the language of printed documents (PDFs), of the e-mails the system sends and of the messages returned by the server, so a customer receives an invoice or a reminder in the language of your business, whatever language the person who sent it uses on screen. A change in Settings reaches the papers and e-mails within seconds.
Time zone	Used for shift times, scheduled reports and daily jobs.
Date format	How dates show on screen and documents.
Max discount (%)	The highest discount percentage allowed on any single sale line, store-wide. Someone with the *Override the store-wide discount cap* permission can exceed it.
Discount	**Off**: no discount field on quotations, sale orders, invoices, purchase orders and bills. **Per line**: each line has its own discount. **Global**: one discount for the whole document. Only one mode applies at a time, store-wide. The POS terminal's own discount is separate and always available (subject to the till rules in chapter 8).

> **Note:** If you choose the Rwanda or Chad chart of accounts while the currency is still the untouched default, the currency is set to RWF or XAF for you. In the same way the business language becomes **Français** for the Chad chart while it is still the untouched English default.

5.3 Taxes

Settings > Taxes lists your tax rates (Code, Name, Rate, Pricing, Status). Press **New tax rate** to add one.

- **Pricing** is either *Tax-inclusive* (the price already contains the tax) or *Tax added on top*.
- Each product picks its own tax rate, or none for tax-exempt goods, in **Inventory > Products**. Any number of rates can be active at once.
- Set a rate to **Inactive** instead of deleting it when it is no longer used; old documents keep it.

5.4 Payment methods

Settings > Payment methods lists the ways customers pay (Order, Code, Name, Cash-equivalent, Store credit, Status). Press **New payment method** to add one.

- Flag **Cash-equivalent** only for methods that should count as *cash in the drawer* when a till is reconciled. Everything else is treated as non-cash at shift close.
- **Order** decides the sequence of the methods on the checkout screen (lower comes first). Set it on the edit screen.
- **Store credit** marks a method that draws on a customer's credit balance.
- **Room charge** ("Charge to room") appears once the Hotel module is active.
- Common choices: Cash, Card, Mobile money, Bank transfer, Cheque, Gift card, Loyalty.

5.5 Branches

Inventory > Branches (or the Branches list in your menu) lists your branches. A branch owns one or more warehouses; sales, purchases and expenses are tagged with their branch so that you can report per branch. Everything recorded before branches existed belongs to the **Head office** (marked with a star), which cannot be deactivated.

Press **New branch** and fill in:

- **Code** (for example KLA), **Name**, optional **Address**, **Phone**, **Email**.
- **Receipt header** (optional). Stored per branch.
- **Tax & numbering** (optional): **Tax ID / TIN**, **EFRIS branch code**, **Document prefix** (for example KLA-). These are stored for each branch.
- **Active**.

5.6 Warehouses and locations

Inventory > Warehouses. A warehouse is a place where stock is kept. Fields: **Name**, optional **Address**, the **Branch** it belongs to (shown when you have more than one branch), and **Low-stock threshold**.

- **Low-stock threshold**: a product is flagged low (on the dashboard and the stock-on-hand report), and an alert e-mail is sent, when this warehouse's stock of it drops below the number. Leave it blank to use the store-wide default (set under Inventory settings).
- Your primary warehouse cannot be deactivated.

- A warehouse can have **Locations**: shelves, bins, a receiving area. Each location has a type that drives behaviour:

Type	Purpose
Receiving	Every purchase or receipt lands on the warehouse's one Receiving location.
Sales	A sale or return targets a Sales location. Every warehouse always has at least one active.
Storage	A plain holding area.
Damaged	Stock is tracked but never counted as sellable.

A location's type cannot be changed after it is created. A register can be pinned to one specific Sales location (chapter 8), and stock transfers can move stock directly between any two locations.

5.7 Apps (EFRIS)

Settings > Apps lists integrations such as the EFRIS (tax authority) connection for Uganda, Rwanda and Chad. Open an app and fill in its settings. Sales sent to EFRIS keep their own fiscal QR code on the receipt.

6 Accounting set-up

Accounting is a core module. Every sale, purchase, expense, payroll run, asset and stock movement posts to it, so it is worth setting up carefully before real transactions begin.

6.1 Choosing the chart of accounts

The first time Accounting is used, a short wizard (the chart-of-accounts country screen) asks which standard chart to start from. The chart sets the account names and numbers and, for Rwanda (RWF) and Chad (XAF), the currency. Choose the chart that matches your country and accountant. After the wizard, **Accounting > Chart of accounts** lets you add, rename and deactivate accounts.

> **Tip:** Agree the chart with your accountant before the first posting. Accounts that already have postings cannot simply be deleted.

6.2 Cost centers

Accounting > Cost centers are optional tags (a department, a project, a branch) that expenses, bills and journal lines can carry, so that you can report cost per center. Create them before you start entering expenses if you want per-center budgets.

6.3 Journal entries

Accounting > Journal entries is for manual postings: accruals, corrections, opening entries. Most entries are posted automatically by the modules; manual journals should be rare and carry a clear description.

6.4 Controls

Accounting > Controls holds the period controls an accountant uses:

- **Close books**: lock a period so that nothing can be posted into it any more. Use it after month-end or year-end review.
- **Bank reconciliation**: match the bank statement against the ledger. The screen shows at most the latest 2,000 ledger lines.
- **Ledger opening balances**: enter the opening balance of each account when you start using the system part-way through a financial year.

6.5 Opening balances (starting on a running business)

1. Decide a cut-over date and close the old system for that date.
2. Import **opening stock** (Inventory) and **customer and supplier balances** with the importers in **Settings > Backup & Data** (chapter 10). Customer and supplier opening balances become real `OB-` invoices and bills, so they age and can be collected and paid like any other.
3. Enter **ledger opening balances** under Accounting > Controls for the accounts that are not covered by the importers (bank, equity, fixed assets and similar).
4. Check the trial balance against your old system before the first live day.

6.6 Incomes

Accounting > Incomes records money received that is not a sale (interest, grants). **Expenses** have their own module with approval rules, VAT handling and budgets, described in the User Manual.

6.7 Year-end

At year-end: review the reports, post any accruals and depreciation (assets depreciate monthly), use **Close books** for the closed period, and take a **Year-end archive** export (chapter 10).

7 Module settings

Every module that needs choices keeps them on its own **Settings** page, at the bottom of that module's menu. All of these pages are **Admin only**. This chapter lists them, in the order of the menu, and explains the choices that matter. Press **Save** on each page; most show *Saved* when done.

> **Tip:** Set a module up *before* staff start using it. Some settings (for example Help Desk deadlines) only apply to records created after you switch them on.

7.1 Sales settings (Sales > Settings)

- **Features**: switch on only what you use: **Promotions** (discounts, buy-X-get-Y, coupons), **Gift cards**, **Loyalty points** (the earning and spending rules are set on the Loyalty page). When a feature is off, its menu entries and till parts are hidden and the server refuses to use it. What was already saved is kept and works again when you switch it back on.
- **Invoice receiver emails**: one address per line. Whenever an invoice is issued, a PDF copy is e-mailed to the customer's own address and to everyone on this list. It is sent only if e-mail is set up (chapter 9); a failed send never blocks or undoes the invoice.

7.2 POS settings (POS > POS settings)

The rules the till enforces. They are the main control against loss at the counter.

Group	Setting
Cash at the till	A closing difference up to a set amount is accepted without comment; a bigger one needs a note saying why. Closing a shift needs the notes and coins counted (option). A paid-out from the drawer above an amount needs a manager's PIN (0 = no limit). Remind the cashier to drop cash into the safe when the drawer holds more than an amount (0 = no reminder).
Voids and returns	Voiding a sale needs a manager (their sign-in or PIN). A sale can be voided only while its shift is open (afterwards it is a return). A return needs a reason. Returns of sales older than a number of days need a manager (0 = no limit). A refund above an amount needs a manager (0 = no limit). The reasons offered for a return (separated by commas).
Discounts and prices at the till	A cashier may give a discount up to a percent alone; above it a manager approves (0 = every discount needs a manager). A discount needs a reason. A manager may change a price at the till. Refuse a price below the item's cost. The reasons offered for a discount.
Receipts	Paper: 80 mm or 58 mm thermal roll. Extra lines under the business details (opening hours, phone, VAT note). Text at the bottom of every receipt. Print the business logo. Print a QR code of the invoice number (sales sent to EFRIS keep

Group	Setting
	their own QR code).

The first printing of a receipt is the original; every later printing is marked **COPY 1, COPY 2…** and listed in **Voids & returns**. "A manager" means an Admin, or a person with the *Approve voids and returns / price changes / big paid-outs* permissions from chapter 4.

7.3 Inventory settings (Inventory > Inventory settings)

- **Allow negative stock** (default on): a sale or manual adjustment may take stock below zero. Untick it to make checkout refuse a sale that would oversell and Stock adjustments refuse a correction that would go negative.
- **Product packaging & breakdown**: lets you break one product into smaller sellable units (a bottle into shots, a case into bottles). Off by default; packaging rules can be configured but a breakdown is refused until this is on.
- **Multi-warehouse**: off by default, the store stays on one warehouse ("Main"). When on, you can create further warehouses, but no sale may complete against Main any more: every register must be assigned a different warehouse (**POS > Registers & Terminal**) before it can ring a sale. Purchases always land on Main's default location.
- **Costing method**: *Weighted average* (recommended), *Last purchase cost*, or *Manual* (never change cost automatically). Landed cost on receiving needs weighted average or last purchase.
- **Daily stock jobs**: *Email a daily low-stock digest* (one e-mail a day to the notifications address, nothing on a day when nothing is low); *Draft purchase requisitions automatically* (each day one DRAFT requisition for products at or below their reorder level; it never submits or approves, and skips products already on an open requisition); *Run from* (store time); and a *Run now* button that uses the saved settings.
- **Restaurant kitchen location**: where restaurant orders and wastage take their stock from (shown when the store has more than one location).

7.4 Expenses settings (Expenses > Settings)

- **Approval limit**: expenses above this amount can only be approved by an Admin or someone with *Approve large expenses*. 0 = no limit.
- **Nobody approves, rejects or voids an expense they entered themselves** (option).
- **Block approval when it would go over the monthly budget**: an Admin can still approve with an override, which is written to the expense history.
- **Days until a bill is due** (used when an unpaid bill is entered without a due date) and **Warn this many days before the due date**.

7.5 Assets settings (Assets > Settings)

- **Warn before a warranty or insurance ends (days)**.
- **"Fully depreciated soon" means within (months)**.
- **Disposal approval limit (book value)**: disposing of an asset whose book value is above this needs an Admin or a user with *Approve large asset disposals*. 0 = no limit.

7.6 Installments settings (Installments > Settings)

- **Email reminders**: send reminders to customers (needs e-mail), how many days before a due date, and how often to chase overdue instalments.
- **Starting values for a new plan**: whole amounts (no cents) for currencies such as RWF and XAF, yearly interest %, grace days, late fee (fixed) and late fee (% of unpaid).
- **Early payoff and default**: the percent of interest waived on an early payoff (changeable per payoff); mark a plan defaulted after this many days overdue (0 = never automatically).
- **Agreement wording**: one term per line, printed under "Terms" on the agreement PDF. Leave empty for the standard wording; have your own text checked by a lawyer.

7.7 Subscriptions job settings (Subscriptions > Job settings)

- **Run from (store time)**: the daily job time, using the time zone in System settings. If the server was off then, it runs when next up the same day.
- **Bill due subscriptions automatically**: each active subscription whose next billing date has arrived gets its invoice. Turn off to bill only by hand with **Bill now**.
- **Email upcoming-billing reminders**: how many days before, to the customer and/or the notifications e-mail.
- **When an invoice stays unpaid**: do nothing, pause the subscription, or cancel it after a number of days past the invoice date. Pausing can be undone with **Resume**; cancelling cannot.
- **Email a daily results digest** to the notifications e-mail (not sent on days when nothing happened).
- **Overdue reminders and catching up**: e-mail the customer when a subscription invoice is overdue (first reminder after n days, then every n days, at most n reminders), and **Missed billing cycles to bill in one run** (1 = one per day; a higher number catches up at once, each with its own period).
- **Run now** uses the saved settings and counts as today's run.

7.8 Help Desk settings (Help Desk > Settings)

- **Public website**: when on, anyone can send a support request and follow a ticket at `/helpdesk` on your site without signing in; options let customers reply and attach files and rate the help. Switch it off to close the page.
- **Notification emails**: one address per line; they get an e-mail for each new ticket, for customer replies to an unassigned ticket, and for missed deadlines.
- **E-mail each reply to the customer** when the ticket has an e-mail address.
- **Deadlines (SLA)**: give every new ticket a first-reply and a resolution deadline by priority (Urgent, High, Medium, Low), counted in calendar hours from creation; time *waiting for the customer* does not count. Only tickets created after you switch this on get deadlines. Options: internal tickets too; warning e-mail before the deadline and alert when missed; **Close a resolved ticket automatically after (days)**.

7.9 Time tracking settings (Time tracking > Settings)

- A manager must approve submitted time (switch off to approve everything automatically).
- **Round stopped timers up to** exact minutes, 5, 6, 10, 15, 30 minutes or 1 hour.

- **Default hourly rate**, **most hours one person can log in a day**, **working hours in a day** (for utilisation).
- **Keeping timesheets tidy**: stop a timer running more than n hours (kept as a draft for the person to check; 0 = never); **Lock time up to and including** a date so approved and invoiced weeks stay as they are; a weekly reminder e-mail to people who have not handed in last week's time; a warning when a project has used n% of its budget (recorded in the audit log).
- **Service product billed time is invoiced under**: the product named "Professional services (time)" can be created automatically; its tax rate and accounting follow the product.

7.10 Booking settings (Booking > Settings)

- **Opening hours**: opens, closes, time between start times, working days.
- **Booking rules**: earliest booking (minutes ahead), furthest ahead (days), hold for payment (minutes), online cancel until (hours before).
- **Reminders and moving a booking**: e-mail the guest a reminder (with a calendar file) n hours before; guests may move their own booking on the public page.
- **Public website**: guests pick a service and a time themselves at `/book`. Also confirmation e-mail, bookings per phone number per day, page title, intro text, and how to pay (shown to guests who owe a deposit).

7.11 Work orders settings (Work orders)

- **New customer jobs need the customer's approval**: work cannot start until an estimate has been sent and approved (can be switched off for a single job).
- **E-mail the customer when a job is completed**: a completion note (PDF) is sent. If the e-mail fails, the job is still completed.

7.12 Attendance and leave settings (HR > Attendance > Settings)

- **Working time**: work starts at, grace period (minutes), hours in a normal day, overtime pays (multiple of normal hourly pay), working days. Clocking in after the start time plus grace counts as late.
- **Leave types**: name, days a year, whether days become available all at the start of the year or are earned month by month, carry-over (max days), paid or unpaid, paid out when leaving.
- **Public holidays**: add the holidays of each year; they are not counted as leave or working days.

7.13 Hotel settings (Hotel > Settings)

- **How room revenue reaches Accounting**: *Every night* (posted by the night audit; best for most hotels), *At check-out*, or *At booking and check-out* (booked value is held as unearned revenue until check-out). The mode can only be changed when no booking is open.
- **Tax on rooms and services (%)**, **check-in time**, **check-out time**, **night audit runs at**.
- **Cancellation and fee policy**: free cancellation until n days before arrival; late-cancellation, no-show, early-departure fees (in nights); late check-out fee. The front desk can waive any fee. See also **Online booking** settings for the hotel's public booking page.

7.14 Restaurant settings (Restaurant > Settings)

Service charge by order type (dine-in, takeaway, delivery) and VAT on it; the most a waiter may discount without a manager; tips (quick-choice percents); quick kitchen notes; usual preparation time and amber/red thresholds for kitchen tickets; rounding of the bill; and table bookings from the public site at `/reserve`.

7.15 Online Store settings (Online Store > Shop settings)

The Online Store needs the **Online Store** module in the licence and activated. Its pages are `/shop` (the public shop) and `/shop/order/<code>` (where a customer follows an order); staff use **Online Store > Online orders** and **Shop settings**. Everything starts closed: the public pages answer "not available" until the module is activated and **Shop is open to the public** is ticked.

Set up, in this order:

1. **Products on the shop**: only products added here are shown or sold. The picture is the product's own; the price is the normal price unless an **Online price** is set. Prices are shown with tax included.
2. **Delivery areas** with a fee and an optional minimum order each. With none, the shop delivers anywhere for free.
3. **Service product for the delivery fee**: a *service* product (no stock) that carries the delivery fee on the sale order and the invoice. Without it, an order that has a delivery fee cannot be accepted. Give it no tax, or tax included, so that the invoice total matches what the customer saw.
4. The shop switches and limits: **Delivery**, **Pick-up**, **Pay cash on delivery or at pick-up**, **Minimum order**, **Free delivery from**, **Units kept back for the till** (the shop sells this many fewer than you hold), **Orders waiting per phone** and **Orders per connection per hour**.

Accepting an order creates a confirmed sale order for the customer (found by phone number, or created), which holds the stock. Completing it fulfils the sale order from the chosen warehouse location (required when several warehouses are in use), issues the invoice and records the payment with the first cash payment method. Staff need the **Operations** area to answer orders.

Behind a reverse proxy (nginx, IIS, a cloud load balancer): add `server.forward-headers-strategy=native` to the server's settings so that the limit per connection sees each visitor's own address. Without it all visitors share the proxy's address and the limit applies to everybody together.

7.16 Law firm settings (Law firm > Settings)

Practice areas and jurisdictions (one per line); "due soon" window; default billing basis for new matters (Hourly, Fixed fee, Retainer, Contingency, Pro bono); daily e-mail of deadlines and tasks to each lawyer; new matters restricted by default; **limitation date warnings** (first warning n days before, a second 3 days before, and one if the date has passed); and **unpaid fees**: the usual payment term and automatic client reminders listing late invoices with a statement attached.

7.17 Real Estate settings (Real Estate > Settings)

Areas you work in; how long a viewing lasts; when a lead is "quiet"; a morning e-mail to each agent about late and due calls and quiet leads (agents need an e-mail address on their user account); and viewing reminders to the person viewing and the agent.

7.18 Property Management settings (Property management > Settings)

Areas; when a lease is "ending soon"; the day rent usually falls due; the usual management fee % and VAT on it; rent invoicing (how many days before the due date, and automatically every morning); and **owner statements every month** (make last month's statements automatically from a chosen day, and optionally agree them and e-mail owners when they have no warnings). There is also a separate reminder and late-fee page for tenants.

7.19 Settings that are not in a module

Settings > System settings, Business details, Email settings, Payment methods, Taxes, Scheduled reports, Apps, Backup & Data are covered in chapters 5, 9 and 10. The **Loyalty**, **Promotions**, **Gift cards** and **Price lists** pages are maintained by staff with the Sales area and are described in the User Manual.

8 Tills, printers and devices

8.1 Registers

A **register** is a till. Create and edit registers under **POS > Registers & Terminal** (Admin only). Each register has a name, can be pinned to one specific *Sales* location of a warehouse (and, when multi-warehouse is on, must be assigned a warehouse other than Main), and can be tied to one device (below).

Cashiers do not need to be given register access one by one: a shift is opened on a register, and the terminal always asks for a **PIN** before it opens. The PIN identifies the cashier and starts or resumes their own working session on that shift.

8.2 Binding a register to one device (version 1.13)

You can tie a register to **one known device** so that it can be used only from that PC.

1. Sit at the till PC and sign in as an Admin.
2. Go to **POS > Registers & Terminal** and press **Bind to this device** on the register.
3. Give the PC a name, and choose whether its network address must also match (the *IP check* option).
4. From then on, any other device is refused by the server for that register: the terminal, sales, opening a shift, offline sync and cash movements, each with a clear message.

Press **Unbind** (an Admin, from any device) to free it. Reports, receipts, history and closing a shift stay available from the back office. A device can be the till of one register only. Registers that are not bound behave as before. Binding and unbinding are written to the audit log (*REGISTER_DEVICE_BOUND* and *REGISTER_DEVICE_UNBOUND*).

If the server sits behind a reverse proxy and you use the IP check, set `app.device-binding.trusted-proxies` to the proxy's address so that the real address of the till is seen.

8.3 Thermal printing and the cash drawer (version 1.12)

Thermal printing is configured **per PC, in the browser**, at **POS > Devices**, because each till has its own hardware. Nothing is stored on the server. The complete guide is in the file `THERMAL-PRINTING.md` supplied with the system; this is a summary.

8.3.1 Connection types

Connection	Use it when	Needs
Windows driver	You want the normal print dialog, laid out for the roll (exactly 80 mm or 58 mm, no margins).	Any browser, the printer's Windows driver.
Serial / COM	You want raw ESC/POS: native text, logo, QR code, cut, cash drawer.	Chrome or Edge on **https** or **localhost**; a printer with a serial port or a virtual COM port.
USB	Same raw printing over WebUSB.	Chrome or Edge; the printer on the WinUSB driver (for example with

Connection	Use it when	Needs
		Zadig).

Chrome or Edge ask once to allow each device (**Connect device…**); the permission is remembered for the site. If a direct printer does not answer, the document is printed through the Windows driver and a red line explains why.

8.3.2 Setting up a till, step by step

1. Open **POS > Devices**.
2. **Receipt printer**: choose the connection, press **Connect device…**, pick the printer. For COM, choose the printer's baud rate (the printer's self-test page shows it).
3. **Paper and options**: paper width (80 or 58 mm), characters per line, code page (use *Plain ASCII* if strange characters print), cut after each document, open the drawer on cash sales.
4. Press **Print test page** and **Open cash drawer**.
5. Press **Save**.

8.3.3 Restaurant printers

The **Kitchen printer** and **Bar printer** are optional. In *Stations on this printer*, type the station names exactly as on the menu items (for example `Grill, Cold` for the kitchen and `Bar` for the bar). Tick *Print kitchen / bar tickets when items are sent to the kitchen*. On the order screen, **Bill** prints the guest's bill (never opens the drawer) and **Tickets** reprints tickets still in the kitchen, marked REPRINT.

8.3.4 What prints where

Document	Printer
Sale receipt and its COPY reprints, return receipt, restaurant bill and paid receipt, X and Z reports	Receipt
Kitchen and bar tickets	Kitchen / Bar
Cash drawer pulse (cash sale, restaurant cash bill, *Open drawer* on the till)	Receipt printer

A4 reports and documents print through the browser.

8.3.5 Customer display on a COM port

Under *Customer display (COM)* tick *Use a customer display on this PC*, press *Connect COM port…*, choose the command set (**ESC/POS**, **CD5220** or **Plain text**), baud rate, data bits, parity and stop bits as set on the display (usual: 9600, 8, none, 1), characters per line (20 for a 2 x 20 display), and welcome lines. Press *Send test message*. On the till the display shows the item just added, then the running total, then PAID with THANK YOU or CHANGE, and the welcome text again.

There is also a second-screen **Customer display** on the till (a browser window) that shows the current cart and total to the customer.

8.4 Settings made on each PC

Because device settings live in the browser, they must be repeated on each till PC and are lost if the browser's site data is cleared. Keep a note of each PC's printer settings.

9 E-mail, alerts and scheduled reports

9.1 E-mail settings

Many features send e-mail: invoices, statements, reminders, daily digests, scheduled reports, forgotten-password links, public-page confirmations. They all depend on **Settings > Email settings**. Fill in:

- **SMTP host** and **port**: 587 (STARTTLS) or 465 (SSL).
- **Username** and **Password** of the mail account.
- **From email** and **From name**: what recipients see as the sender.
- **Skip certificate check**: only for a mail server with a self-signed certificate; avoid it when you can.
- **Business alerts email** (the "notifications email"): the address that receives system alerts, daily digests, low-stock e-mails and so on.

Press **Send test email** and check that it arrives. Many features show a message such as "Needs e-mail set up in Settings" until this works.

> **Tip:** For Gmail or Microsoft 365 accounts you normally need an *app password* rather than the person's normal password. Use a dedicated mailbox for the system, not a personal one.

A failed e-mail never blocks the underlying work: an invoice is still issued, a job still completed. Look at the server log (chapter 11) if mail does not arrive.

9.2 Scheduled reports

Settings > Scheduled reports e-mails a report as an **Excel or PDF** attachment on a schedule.

1. Press to add a schedule and give it a **Name** (for example *Monday stock valuation*).
2. Choose the **Report** and the **Format** (Excel .xlsx or PDF).
3. Choose **Repeat**: Daily, Weekly (pick the weekday) or Monthly (pick the day of month), and the **time**.
4. Add optional filters (Category, Warehouse, Branch, Days) where the report supports them.
5. **Send to**: comma-separated addresses, or blank to use the notifications e-mail.
6. Tick **Active** and save.

How periods are chosen: reports that cover a date range send the period that **just ended**: yesterday for daily, the previous 7 days for weekly, the previous calendar month for monthly. Snapshot reports (stock valuation, ageing and similar) are taken at the moment of sending.

The list shows each schedule's last run. **Send now** sends it immediately; a paused schedule is marked *(paused)*. More than fifteen reports are available, including the asset reports added in version 1.22.

> **Note:** Scheduled reports are not stopped when the report's module is locked by an expired licence.

9.3 Other automatic e-mails and daily jobs

Job	Where it is switched on
Low-stock digest, automatic draft purchase requisitions	Inventory settings

Job	Where it is switched on
Customer collection reminders (ageing)	Customers > Collections
Subscription billing, reminders and digest	Subscriptions job settings
Rent invoicing, tenant reminders, owner statements	Property management settings
Lead follow-ups, viewing reminders	Real Estate settings
Limitation-date warnings, fee reminders, daily deadlines	Law firm settings
Installment reminders	Installments settings
Booking and appointment reminders	Booking settings
Help Desk deadline warnings and automatic closing	Help Desk settings
Weekly timesheet reminder	Time tracking settings

The jobs run once a day at the time shown in each module's settings, in the time zone of **System settings**. If the server was off at that time, a job runs when the server is next up the same day.

10 Backup, restore, import and export

All of this is on **Settings > Backup & Data** (Admin only).

10.1 Backups

- A **full database backup runs automatically every night**. You can also press **Backup now** at any time.
- The list shows each backup file with its **size** and **creation time**, with **Download**, **Restore** and **Delete** buttons.
- **Where the files are:** when the server runs from its jar (for example as a Windows service), backups are written to a folder named **backups** next to the jar, beside the **logs** folder; the folder is created if it is missing. To use another folder, set `app.backup.directory` in `application.properties`, as an environment variable (`APP_BACKUP_DIRECTORY`) or on the command line.
- Backups include the Flyway migration history, so a restore brings the data and its structure version back together.

> **Warning:** The backup lives on the server. If the server is lost, so are the backups. Regularly **Download** a backup and copy it to another computer or cloud storage, together with the server's document storage folder.

10.1.1 Restoring

1. Take a fresh backup of the current state first (**Backup now**), in case you restore the wrong file.
2. Press **Restore** next to the backup you want, or use *Restore from an uploaded backup file* for a file you downloaded earlier or brought from another machine.
3. Confirm. **Restoring replaces the entire current database.**
4. While it runs the whole system goes into **maintenance mode**: everyone else (cashiers, other admins) sees a "please wait" screen.
5. When it finishes, **restart the application** for a clean start.

> **Warning:** Restoring a backup made by a *newer* version into an *older* version makes the migration check fail ("a migration it does not know"). Deploy the newer version instead, then restore.

10.1.2 Good practice

- Test a restore on a spare computer at least once, before you need it.
- Keep at least one backup per week for a month and one per month for a year, outside the server.
- Back up before every upgrade (chapter 12), every import, and before closing the books.

10.2 Bulk import

Bulk import loads data from a CSV or Excel file. The importers include products, customers, suppliers, categories, brands, **customer balances**, **supplier balances** and **ledger balances**, plus module imports (employees, law clients and matters, real estate, and others) inside their own modules.

The process is always the same and is designed to be safe:

1. Press **Download template** for the importer you want and fill it in. Keep the header row.

2. Choose the file and press **Check file** first. It processes the file exactly as the real import would, but **saves nothing**.
3. Read the result: how many rows were read, how many **can be imported** and how many **will be skipped**, a list of **Rows that will be skipped** with the reason for each, and a preview of **what will be created**.
4. Correct the file and check again until the result is what you want. A row that cannot be used is skipped and listed with the reason; the rest of the file still imports.
5. Press **Import**. The result shows how many were imported and skipped.

Notes:

- **Opening balances** (customer and supplier) become real OB- invoices and bills, so they age, appear in statements and collections, and are paid like any other.
- For **ledger balances**, you can choose *Put any difference in Owner's Equity* so that the opening journal balances.
- Module imports read the first sheet only, up to 2,000 rows (.xlsx or CSV).
- Imports have not been tested on very large files; split big files.

10.3 Export and year-end archive

- **Export products** and **Export customers** download the current lists.
- **Year-end archive** downloads a snapshot of one year's business activity (sales, purchases, payments, stock adjustments) for closing out a financial year.

> **Note:** The year-end archive is a curated snapshot, **not** a full database backup. Always keep a real backup as well.

Each list in the system also has its own **Excel, PDF and CSV** export, with the same filters as the screen.

11 Logs and monitoring

The system keeps three records of what happens. They answer different questions.

Record	Question it answers	Where
Audit Log	Who changed what in the business?	Settings > Audit Log
Sign-in log	Who signed in, and who failed to?	Settings > Sign-in log
Server logs	What is the application itself doing, and why did something fail?	Settings > Server logs

All three are Admin only.

11.1 Audit Log

Every important action is recorded with the time, the user, an action code and a description. Examples: voids and returns, stock adjustments and their approvals, price and discount overrides, imports and exports, shift opening and closing, binding a register, role and staff changes, time-tracking budget warnings, and many module actions.

Use the filters at the top: a **search box** (user or description), an **action** drop-down (*All actions*), **From** and **To** dates and a page size (25, 50, 100, 200). The table shows **When, Who, Action, What happened**.

Typical uses: find out who approved a discount, who voided a sale, who changed a price, who exported the customer list.

11.2 Sign-in log

Every sign-in attempt is recorded, successful or not. The top of the page shows how many people **signed in in the last 24 hours**, how many **failed attempts** there were, and how many **addresses had failures**. Below, filter by *User name*, by result (*All attempts*, *Signed in*, *Failed*) and by dates. Columns: **When, User name, Result, Why, Address, Device**.

Watch for many failures from one address or for one user name (a sign someone is guessing passwords). Press **Refresh** to update.

11.3 Server logs

The server writes rotating log files. When the backend runs as a jar, for example as a service, they are in a folder named **logs** next to the jar (before version 1.26.2 they were in the jar's folder itself). When it is started from a development tool without a jar, they are in a `logs` folder where it was started. The setting `app.logging.dir` (or the environment variable `APP_LOGGING_DIR`) can point them to another folder, and then it wins.

The files are:

- `niwe-erp.log`: the normal application log.
- `niwe-erp-error.log`: warnings and errors only.
- `niwe-access.log`: one line per request.

Every line carries a **request id** and the **user**. The same request id is returned to the browser in the `X-Request-Id` header, so a support person can match what a user saw with the server's record.

On **Settings > Server logs**:

- Choose a file, a level (*All levels*, *Info and above*, *Warnings and errors*, *Errors only*), type a text to search, and pick how many lines to show (last 100, 300, 1000, 2000).
- **Auto-refresh** keeps watching. **Download file** gets the whole file (the screen shows only the last part of a large file).
- Before the server has written anything the page says *No log files yet*.
- Only files whose names start with `niwe-` and end with `.log` appear here. Other files in the same folder are never shown or downloadable from the screen.

When you ask NIWE for support, send the request id and the relevant part of `niwe-erp-error.log`.

The log behaviour is controlled by the `app.logging.*` properties in the backend settings (folder, size and number of files to keep). The account that runs the service must be allowed to write in the log folder; if it is not, no log files appear.

> **Tip:** After installing the backend as a service, start it once, then open **Settings > Server logs** to confirm the files appear. If the page says *No log files yet*, check that the service account can write in the jar's folder (it has to create the `logs` folder).

11.4 Other places to look

- The **dashboard** has a *Needs your attention* section (overdue invoices, low stock, pending approvals and similar).
- **Settings > Modules** shows the licence status and expiry.
- Each module's **Overview** and **Reports** pages show its day-to-day health.

12 Upgrades and migrations

12.1 How the database stays in step

The database structure is owned by **Flyway**. Each release carries numbered SQL migration files; on start-up Flyway applies any the database has not seen and records them in the `flyway_schema_history` table. Hibernate only *validates* that the code and the database agree (`ddl-auto=validate`); it never changes tables.

Rules that matter to an administrator:

1. **Never edit or delete a migration that has been applied.** Flyway stores a checksum and refuses to start if a file changed.
2. A new release is simply started: the new migrations run once, each in its own transaction.
3. **Back up first.** Always (chapter 10).

12.2 Installing a new release

Each release is delivered as a package (for example `niwe-pos-v1.22.0.tgz`) with its own release notes (`RELEASE-NOTES-v1.22.0.md`). The usual steps:

1. Announce a short maintenance window. Take a **Backup now** and download it.
2. Stop the backend.
3. Unpack the package over the project folder.
4. Run the release's apply script(s), as named in the release notes: `tools/apply-vXXXX.py` and `tools/wire-vXXX.py frontend/src/app`.
5. Build the frontend and the backend.
6. Start the backend. Watch the log for Flyway applying the new migrations and for *Started PosApplication*.
7. Open the system, sign in, and check **Settings > Modules** (licence and module states) and a few screens.

If a release adds a new permission (for example *Approve large asset disposals*), nobody has it until you grant it under **Staff**; Admins have all permissions automatically.

12.3 The first time on an old database (one time per database)

The current builds start with schema validation. If an old database was created before Flyway and lacks tables or columns:

1. Take a backup.
2. Start the backend **once** with the extra profile `dbsync` (for example `mvn spring-boot:run -Dspring-boot.run.profiles=dbsync`, or your usual profile plus `dbsync`).
3. Wait for *Started PosApplication*, then stop it.
4. Start normally. Flyway baselines the database and carries on.

> **Warning:** Never use `dbsync` on a database that Flyway already manages.

A database created during development with the older 23 migrations must be dropped and recreated (safe only if it has no real data), because the single V1 baseline has a different checksum.

12.4 Version history in brief

Version	Main theme
1.1	Module licensing
1.2 to 1.4	Property Management, Real Estate and Law firm improvements
1.5	Core: paging, search and exports everywhere; collections; imports with check-first; accounting controls; price lists; promotions, gift cards and loyalty
1.6 to 1.10	HR and Payroll, Hotel, Restaurant, POS, Manufacturing; decimal quantities
1.11	Receipt tax ID, dashboard paging, sales feature switches
1.12	Thermal printing and COM customer display
1.13	Register tied to one device
1.14	Server logs, sign-in log, wider audit log
1.15 to 1.22	Work orders, Time tracking, Booking, Installments, Subscriptions, Help Desk, Expenses, Assets

12.5 If a start-up fails

- *Migration checksum mismatch*: a migration file was changed after being applied. Restore the original file.
- *Schema-validation: missing column/table*: the database is behind the code. Run the one-time `dbsync` start described above (after a backup).
- A message naming a *migration it does not know*: you are running an older build against a database made by a newer one. Deploy the newer build.

13 Security checklist

Work through this list before the system holds real data, and again once a year.

13.1 Before going live

- ☐ **Change the admin password** and remove or deactivate the demo `cashier` account (or change its password and PIN).
- ☐ Change **all terminal PINs** from the demo values (1234 and 5678).
- ☐ Set a long random **`JWT_SECRET`** (the `app.jwt.secret` setting). Anyone who knows it can forge a sign-in.
- ☐ Set **`app.seed-demo-data=false`**.
- ☐ Use your own database user and password (`DB_USER`, `DB_PASSWORD`), never share them in files that are copied around, and do not expose the database port to the network.
- ☐ Serve the system over **https** (a reverse proxy or web server certificate). This is also required for direct thermal printing on till PCs that are not on localhost.
- ☐ Set `app.cors.allowed-origins` to your real address only.
- ☐ Make sure the **`dev` profile** is never used and `app.license.dev-all-modules` is off.
- ☐ Set up **E-mail settings** so that staff can use *Forgot password?*, and give each person an e-mail address.
- ☐ Confirm that a nightly backup exists and that a copy leaves the server.

13.2 Ongoing

- Give each person their own account; use custom roles and the approval permissions rather than extra Admins.
- Review **Staff** quarterly and untick *Active* for leavers immediately.
- Review the **Sign-in log** weekly for repeated failures.
- Review the **Audit Log** for voids, returns, price overrides and stock adjustments.
- Keep the server's operating system, Java, PostgreSQL and the browser up to date.
- Restrict who can reach the server physically and by remote access.
- Bind each register to its till PC (chapter 8) where tills sit in public areas.
- Download a backup before every upgrade and keep one off-site.

13.3 If you open the Online Store to the internet

The public shop is the one part of NIWE ERP meant for strangers, so treat the server as exposed.

- Serve it over **HTTPS only** (a certificate on the web server or load balancer) and keep the admin screens behind the same address or a separate one you can restrict.
- Only `/shop`, `/shop/order/...` and the `/api/public/...` addresses are meant to be public. The staff screens still need a sign-in, and the shop's server calls never return costs, suppliers or other customers.

- Set `server.forward-headers-strategy=native` when a proxy sits in front, so the per-connection limit works (chapter 7).
- Keep **Orders waiting per phone** and **Orders per connection per hour** low, watch the **New** orders tab, and phone a customer before packing an unusual order.
- Customers' names, phone numbers and addresses are personal data: back up daily, keep the backups private, and check what the data-protection rules of your country require before you launch.
- Have the server tested by someone outside the team before the shop is announced.

13.4 What the system does for you

- Passwords and PINs are stored securely, not in clear text.
- Sign-in uses signed tokens that expire; every server request is checked against the person's role, areas and branch access, not just hidden in the menu.
- Locked modules are refused by the server, not only hidden.
- Deleting is limited: most records are voided, reversed or deactivated, leaving a trail in the audit log.

14 Troubleshooting

Problem	Likely cause and what to do
A module is missing from the menu	It is not in the licence, not yet activated, or the person's role has no access to it. Check **Settings > Modules**. Activation is by an Admin and is permanent.
"X is not part of your license" (403)	The module is locked. Install a licence that contains it (chapter 3).
All non-core modules locked suddenly	The licence has expired. Nothing is lost. Install the renewal.
A person cannot see a screen they should	Their custom role has *No access* for its area, the role is *inactive*, or the screen is admin-only (Settings, Staff, Backup). See the area table in chapter 4.
A cashier cannot approve a void/discount	They lack the matching approval permission. Add it under **Extra permissions** on their staff record, or ask an Admin to type their PIN.
"PIN not recognized" at the terminal	The PIN is wrong, or belongs to an inactive person. PINs are 4 to 6 digits and unique among active staff. Reset it on the Staff list.
A till says it is refused / wrong device	The register is bound to another device. Unbind it (Admin) or bind it to this PC (chapter 8).
Cannot sell: warehouse message	Multi-warehouse is on and the register still uses the Main warehouse. Assign a different warehouse to the register.
Sale refused because of stock	*Allow negative stock* is off and the sale would oversell. Receive stock or adjust it.
A thermal printer does nothing	Check the connection type, that the browser is Chrome or Edge on https/localhost, press *Connect device…*, then *Print test page*. A red line on the printout says why it fell back to the driver.
Strange characters on receipts	Change the code page to *Plain ASCII* in the PC's device settings.
No e-mail arrives	Check **Email settings** and use **Send test email**. Check the recipient's address and the server log. A failed e-mail never blocks the work itself.
Daily job did not run	Check the module settings for the job's on/off switch and time, and that the time zone in System settings is right. A job runs when the server is next up that day.
Backup restore stuck on "please wait"	Restore is running. Wait, then restart the application when it finishes.
Server will not start: checksum or schema message	See chapter 12, *If a start-up fails*.
Import skips many rows	Read the reasons in *Rows that will be skipped* after **Check file**; correct and check again.
A report looks wrong	Check the filters and dates, and whether the books were closed for that period (Accounting > Controls).

Problem	Likely cause and what to do
A user forgot their password	If they have an e-mail address on file, use *Forgot password?* on the sign-in screen. Otherwise an Admin sets a new password on the Staff record.

14.1 When you contact support

Send NIWE: the **version number**, the **installation id** (Settings > Modules), what you did, what you expected and what happened, a screenshot, and the **request id** and the matching lines from `niwe-erp-error.log` (Settings > Server logs).